

Day 6 Problem Set

Geometry Summer Credit | Triangle Relationships and Coordinate Triangles

Name:	Date:	Period:
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Goal

Practice triangle inequality, side-angle ranking, midpoint, distance, midsegments, and medians.

Support Practice

1.	Solve $2x + 4 = 18$.
2.	Find the missing angle: 48 degrees, 77 degrees, and x .
3.	Find the midpoint of (2, 6) and (10, 4).
4.	Find the distance between (0, 0) and (6, 8).
5.	A midsegment is half the length of the parallel side. If the parallel side is 20, what is the midsegment?

Core Practice: Triangle Inequality

6.	Can 5, 6, and 8 form a triangle? Explain.
7.	Can 4, 4, and 9 form a triangle? Explain.
8.	Can 7, 10, and 17 form a triangle? Explain.

9. Find the range for x if the side lengths are 6, 9, and x .

10. Find the range for x if the side lengths are 11, 15, and x .

11. A student says 2, 3, and 5 form a triangle because $2 + 3 = 5$. Explain the mistake.

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Continued

Core Practice: Side-Angle Relationships

12. In triangle ABC, angle A = 50 degrees, angle B = 75 degrees, and angle C = 55 degrees. Order the sides from shortest to longest.

13. In triangle DEF, DE = 14, EF = 9, and DF = 12. Order the angles from smallest to largest.

14. A triangle has angles 30 degrees, 65 degrees, and 85 degrees. Which side is longest?

15. A triangle has side lengths 18, 22, and 25. Which angle is largest?

Core Practice: Coordinate Triangle Tools

16. Find the midpoint of A(-4, 9) and B(8, 1).

17. Find the midpoint of C(5, -7) and D(-3, 11).

18. Find the distance between E(-1, 3) and F(7, 9).

19. Find the distance between G(2, -5) and H(10, 10).

20. A midsegment is parallel to a side of length 34. Find the midsegment length.

21. A midsegment is 12.5 units long. Find the length of the parallel side.

22. A median from vertex A meets side BC at its midpoint. If B(-2, 6) and C(8, -4), find the point where the median meets BC.

23. A triangle has vertices P(0, 0), Q(8, 0), and R(2, 6). Find the midpoint of QR.

24. Spiral: Triangle JKL is congruent to triangle MNP. If angle K = $3x + 7$ and angle N = 70 degrees, find x.

25. Challenge: Explain why a midsegment and a median are different, even though both involve midpoints.